

MATING WORM DATA (EXISTING — ENTRY)	
Major diameter da1	28.00 mm
Root diameter df1	19.33 mm
Axial pitch Px	6.350 mm
Starts z1	1
Pitch diameter d1 (derived)	23.96 mm
Lead L = z1·Px	6.35 mm
Lead angle λ	≈ 4.8°
Diameter quotient q = d1/m	11.853

TOOTH DATA (REFERENCE)	
Module m = Px/π	2.021 mm
Pressure angle	20°
Axial pitch Px	6.350 mm
Worm starts z1	1
Wheel teeth z2	150

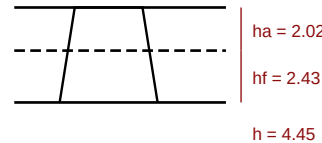
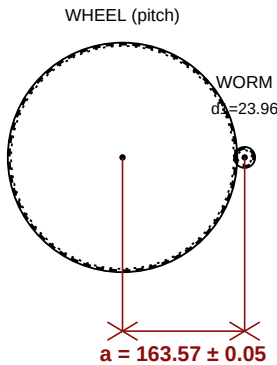
RECOMMENDED BLANK (BEFORE CUTTING)	
Outside diameter	307.2 mm
Face width	35 mm
Bore	60 mm H7
Hub diameter	120 mm
Hub length	30 mm
Material	Bronze / CuSn12

GENERAL NOTES	
1. Pressure angle: 20° (standard). 2. Tooth form generated by mating worm. 3. All dimensions in millimeters. 4. Unspecified fillet radius = 1.5 mm. 5. Deburr all edges 0.2 × 45°. 6. Material: Bronze (CuSn12) or equivalent. 7. Heat treatment (if required) after machining. 8. REV B: center distance uses worm pitch dia. d1.	

KEYWAY DETAIL
Width: 18 P9
Depth: 11 P9
(from top of bore)
18 × 11 × 70 DIN 6885-1

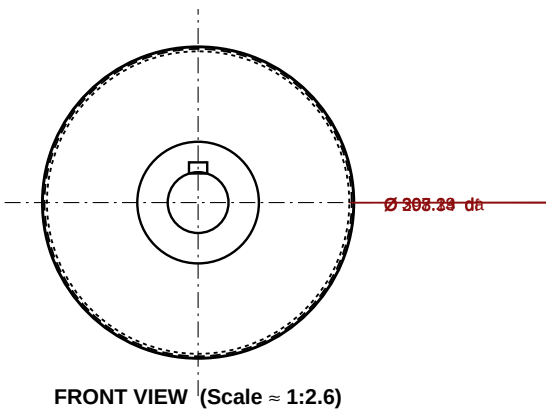
Formulas: $m = Px/\pi$ · $d1 = da1 - 2m$ · $d = m \cdot z2$ · $a = (d1 + d)/2$ · $\lambda = \arctan(L/(\pi \cdot d1))$ · $ha = m$ · $hf = 1.2m$

CENTER DISTANCE (corrected — pitch diameters)

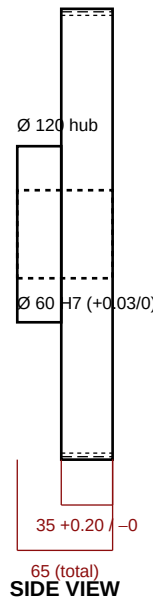


SECTION A-A (tooth depth)

Keyway: 18 × 11 × 70 DIN 6885-1



FRONT VIEW (Scale ≈ 1:2.6)



SIDE VIEW

WORM WHEEL PARAMETERS (CORRECTED)	
Number of teeth z2	150
Module m (from Px)	2.021 mm
Axial pitch Px	6.350 mm
Circular pitch p	6.350 mm
Pitch diameter d	303.19 mm
Outside diameter da	307.23 mm
Root diameter df	298.34 mm
Addendum ha	2.02 mm
Dedendum hf	2.43 mm
Whole depth h	4.45 mm
Tooth thickness s (pitch)	3.175 mm
Center distance a	163.57 mm
Reduction ratio i = z2/z1	150:1
Worm lead L = z1·Px	6.35 mm
Estimated lead angle λ	≈ 4.8°

REVISION B — CALCULATION FIX

Rev A used $a = (da1 + d)/2 = (28.00 + 303.15)/2 = 165.58$ mm (worm MAJOR dia.).
Rev B uses $a = (d1 + d)/2$ with $d1 = da1 - 2m = 23.96$ mm → $a = 163.57$ mm.
 $\Delta a = -2.01$ mm. Lead angle $\lambda = 4.8^\circ$ now consistent with $d1$.

WORM WHEEL (CROWN) — 150 TEETH

Generated by mating worm · Pressure angle 20°

Material: Bronze / CuSn12 or equivalent

Drawing No.

WW-150T-2.0-01

Tolerances: ISO 2768-mK

Projection: Third Angle

Rev

B

Scale

1:2

All dims in mm

Date

2026-07-09